



Facebook challenges Twitch and Youtube

Facebook has announced that it will now feature a tab on its website and app that's dedicated to gaming. <https://digit.in/apr19-75>



Google+ to get backed-up

The Internet Archive is working to preserve public Google+ posts before the service shuts down in April. <https://digit.in/apr19-76>

Corsair Crystal 680X

Expensive and beautiful

Corsair scaled up the tiny Crystal 280X PC case to give us the Crystal 680X.

What looks like a cousin to the Corsair Air 740, the internals are quite similar since it follows the dual-chamber design. In one compartment, the main components attached to the motherboard are installed whereas in the other, your storage devices and PSU is installed. Not only does this make the build process easier, but also contributes to better airflow potential.

Build and design

Corsair has gone with thick gauge steel plates inside for the motherboard and fan mounting plates. However, we found the HDD and SSD trays to be incredibly and inexplicably delicate.

A dual-chamber layout works well to separate the components and give you less headache when you're building a system. You will find dust filters everywhere except on the top that stays open all the time. If the user chooses not to install fans on the top panel, then you don't have any option to close it off. Although you can lower the distance between the top glass panel and the opening, it still isn't enough to keep the dust out. Cable management is implemented well and we were able to quickly build our test rig into it. There are enough tie points to route your cables to every component and enough room to stow away the extra PSU cables.

Features

Three Corsair LL120 RGB intake fans are pre-installed on the front panel that connects to the Corsair Lighting Node PRO allowing illumination and speed control using Corsair's iCUE software. You can change the addressable LEDs on the fans individually



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according to your preferences. Since it's a Crystal series case, there's tempered glass on the top, left side and front. The tinted glass side panel is attached using a hinge mechanism,

so it's easy to access your internal components.

Thermal performance

The Crystal 680X keeps things cool quite brilliantly. After sitting idle for 15 minutes, the Intel Core i7-4960X came down to 34 degrees Celsius while the Radeon HD 7790 settled at 34 degrees. Do note that the case comes with three intake and one exhaust fan, so such low temperatures weren't surprising. When we put the system under load, the CPU reached 51 degrees and the GPU touched 69 degrees. The CPU temperature seemed to be lower than our usual records, so we repeated the tests again and found almost the same with little variance.

To determine whether the case can do better with more fans, we populated the remaining slots that included the bottom panel as intake and top panel for exhaust. While the extra did make some difference to the GPU temperatures, we didn't notice much

of a change in the CPU temperatures. Another observation was that whether we occupied the top panel or not, the temperatures didn't change much considerably. Only notable difference was that the bottom intake fans reduced the maximum temperature of the GPU by three to four degrees.

We can certainly say that you won't really need extra fans for cooling. As for aesthetics, the three front panel fans are already illuminated with RGB lighting. You might want to replace the exhaust with an RGB fan to complete the overall look.

Final thoughts

The Corsair Crystal 680X is expensive. Considering everything it offers including the insane build quality, features and the wonderful thermal performance, the price is still too high to consider. Of course, as an enthusiast builder, you might already be loaded since this isn't a case one would buy for regular or gaming purposes. The inclusion of the Corsair Lighting Node PRO does try to justify the high price to an extent and although it is necessary to illuminate the fans, it automatically turns the case into a smart case where the other fans can also be connected and controlled using iCUE. Hardly any compromise has been made in terms of build quality, except on the storage device trays. Apart from those, the rest of the components inside are sturdy. Because of the dual-chamber layout, we spent less time routing cables and searching for the best loops. It also gives more volume to the motherboard compartment and allows room for extra intake if the existing one isn't enough.

—Abhijit Dey

SPECIFICATIONS

MOTHERBOARD FORM FACTOR: Mini-ITX, Micro-ATX, ATX, E-ATX | POWER SUPPLY STANDARD: ATX standard (max. length of 225mm) | I/O PORTS: 2x USB 3.0, 2x USB 3.1 Gen-2 Type-C, 1x microphone, 1x headphone | DRIVE BAYS: 3 x 3.5-inch HDD, 4 x 2.5-inch SSD | COOLING SUPPORT: 3 x 120mm (front), 2 x 140mm (front), 2 x 120mm (top), 2 x 140mm (top), 1 x 120mm (rear) | RADIATOR SUPPORT: Up to 280/360mm (front), up to 240/280mm (top), up to 120mm (rear) | CLEARANCES: CPU cooler - 180mm, Graphics card - 330mm | DIMENSIONS (H x W x D): 423mm x 344mm x 505mm

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